

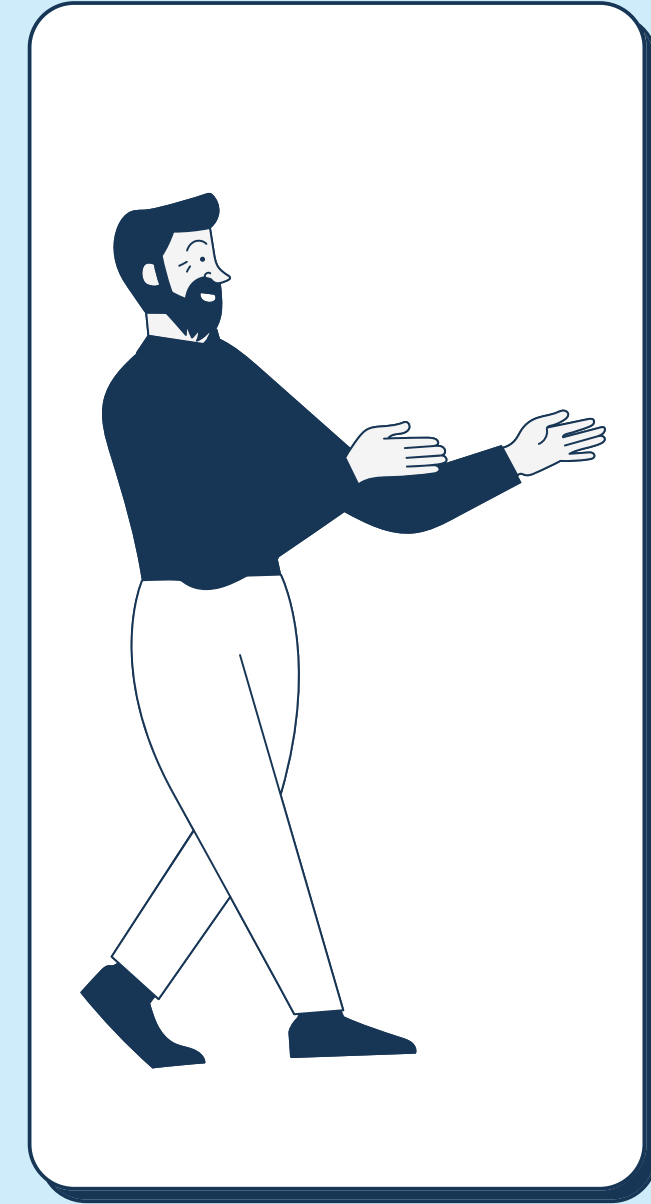
Skill Challenge #14

#dynamicprotocols



Requirements

- > Define two new types: `PlayingCard` and `CardDeck`
- > `PlayingCard` instances should support two instance attributes: suit and rank. The specified attributes should be standardized to lowercase, but not otherwise validated. Equality and a good string representation should be supported.
- > `CardDeck` will represent the traditional deck of 52 cards:
 - 4 suits ('spades', 'diamonds', 'clubs', 'hearts') **x**
 - 13 ranks (2 through 10, inclusive, Ace, Jack, King, Queen)
- > `CardDeck` should be a pythonic sequence supporting the following functionality:
 - membership testing with 'in' keyword (returns bool)
 - square bracket indexing with ints (`PlayingCard` instances should be returned)
 - slicing with slice objects (`CardDeck` should be returned)
 - iteration in for loops
 - concatenation through '+' operator, supported between two instances of `CardDeck`, or `PlayingCard` & `CardDeck`
 - concatenation/extension through '*' operator, supported between an instance of `CardDeck` and int
 - both of the above concatenation operations should be commutative



Requirements Continued

- > If instantiated with no args, `CardDeck` should generate a full deck of 52 cards. If instantiated with a python list of at least one `PlayingCard`, `CardDeck` should reflect only the list of cards passed to it at instantiation. If a python list containing no valid `PlayingCard` instances, the default full deck of 52 cards should be generated.
- > Finally, `CardDeck` should support a `draw(n=1)` method that randomly selects a card from a given deck and returns it without replacement, i.e. the number of cards remaining in the deck decreases by 1
- > If a user draws more than 1 card at a time, e.g. `draw(n=3)`, a `CardDeck` type should be returned; if only one card is drawn, a `PlayingCard` type should be returned.

